greater understanding. An ordinary data table may be a collection of insignificant values, but if we describe each column, and even better, link it to a standardized vocabulary or logical model, this data starts to become meaningful. As its user, I can consciously decide what is relevant to me and what is not, often without even necessarily contacting the original creators.

Of course, this is only an example of metadata related to semantics. However, there can be many more types of metadata.

The following is a list of some of the most relevant data product metadata types. In fact, this is not a closed list:

- *Business metadata*—Describing information about the owner (data product owner), describing the domain context (e.g., abstract, assumptions), domain affiliation, organizational unit, license type, and expiration date
- *Technical metadata*—About the technical team responsible for it, physical access method, links to the data product page (if any), and links to download examples
- *Operational metadata*—Description of ports, description of access policies, and how to obtain extended access
- *Physical schema*—Description of the data product, its attributes, data types, constraints, and relationships to other elements
- *Semantic metadata*—Links of the physical model to standardized vocabularies and logical models
- *Local lineage*—Information about the direct sources of the data from which the data product was created
- *Full lineage*—Information about the overall sequence of links that shows how data was created in a given data product
- *Quality metrics*—Metrics related to data quality, the amount of correctly and incorrectly defined data, information about missing data, incorrect formats, and compliance with data governance rules
- *Operational metrics*—Information about whether the data product is available, number of users, usage statistics, and SLA metrics

5.5.3 Metadata as code